

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 1 (Generated practice paper - Set 13)**

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

01 . 1

Explain why mineral extraction causes more land-use conflict than most other human activities.

[2 marks]

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01 . 2

Describe how each of the following environmental impacts of an operating mine can be reduced.

[5 marks]

Dust

Noise

Turbid drainage water

Acidic leachate from spoil

Loss of amenity

01 . 3

Explain how spoil heaps can be made stable and describe what happened at Aberfan in 1966.

[3 marks]

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10

02 . 1

Define the term **persistence** as applied to a pollutant.

[1 mark]

02 . 2

Explain how the persistence of a pesticide is affected by **two** named environmental factors.

[4 marks]

5

03 . 1

Explain why nitrates in drinking water can cause **blue baby syndrome** (methaemoglobinaemia).

[3 marks]

03 . 2

Explain why the link between nitrates in drinking water and stomach cancer is uncertain.

[2 marks]

0 3 . 3

Describe **three** ways farmers in a Nitrate Vulnerable Zone can reduce nitrate leaching into aquifers.

[3 marks]

1

2

3

0 3 . 4

Explain how phosphates are removed from sewage effluent and from lakes polluted in the past.

[2 marks]

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10

0 4 . 1

Explain why current global energy use is described as unsustainable.

[3 marks]

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0 4 . 2

Describe the environmental impacts of exploiting fossil fuels that occur **before** the fuel is used.

[4 marks]

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0 4 . 3

Explain why an 'energy gap' could occur if society waits until fossil fuels become expensive before developing alternatives.

[3 marks]

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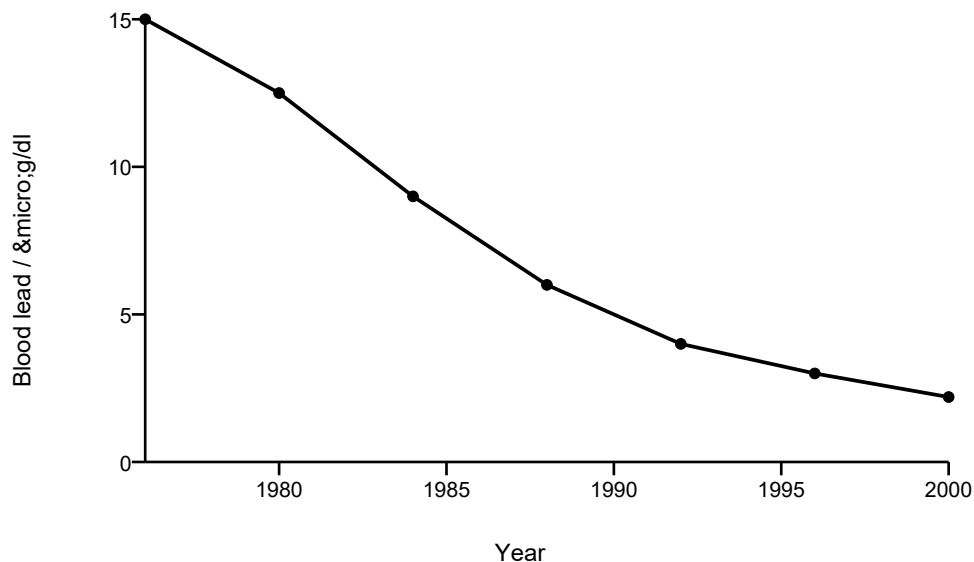
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10

0 5

Figure 1 shows the mean concentration of lead in the blood of children in a large city between 1976 and 2000, and the year in which lead additives in petrol began to be phased out.

Figure 1**0 5****1**

Calculate the percentage decrease in mean blood lead between 1976 and 2000.

[2 marks]**Answer**

%**0 5****2**

Explain why tetraethyl lead was added to petrol and why this caused a public health threat.

[3 marks]

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0 5 . 3

Suggest **two** reasons, other than petrol additives, why blood lead concentrations continued to fall after 1990.

[2 marks]

1

2

0 5 . 4

Explain how adding small amounts of phosphoric acid to a water supply reduces lead in drinking water.

[2 marks]

0 5 . 5

Explain why discarded lead fishing weights and shotgun pellets can poison waterfowl.

[1 mark]**10**

0 6

Mercury released into a lake from a factory was in the inorganic form. Fish in the lake were later found to contain high concentrations of methyl mercury, an organic form.

0 6 . 1

Suggest how the inorganic mercury was converted to methyl mercury.

[2 marks]

0 6 . 2

Explain why methyl mercury is a more hazardous pollutant than inorganic mercury.

[3 marks]

0 6 . 3

Explain why the health risk from eating fish from the lake may remain for many years after the discharges stop.

[2 marks]

0 6 . 4

Suggest **three** factors that should be included in a Critical Pathway Analysis of a proposed mercury discharge.

[3 marks]

1

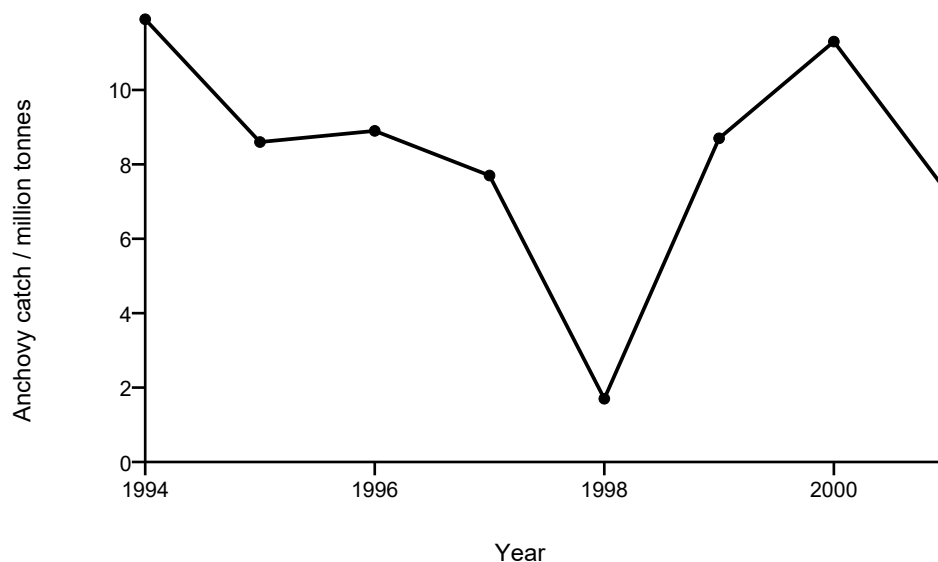
2

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10

07

Figure 1 shows the sea surface temperature anomaly in the eastern Pacific off Peru and the annual Peruvian anchovy catch for a series of years including a strong El Niño event in 1997-98.

Figure 1**07****1**

Describe the normal (non-El Niño) pattern of winds and ocean currents in the tropical Pacific and explain why the Peruvian coast supports a rich fishery.

[4 marks]

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07**2**

Explain how an El Niño event caused the collapse of the anchovy catch in 1998 shown in Figure 1.

[3 marks]

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07 . 3

Describe **two** other effects of El Niño events.

[2 marks]

1

2

07 . 4

Suggest why some scientists link global climate change to changes in the frequency of El Niño events, and why this link is uncertain.

[1 mark]

10

08

A dam and reservoir are to be built on a river to supply water and regulate flow.

08 . 1

Explain **three** effects the reservoir may have on wildlife in and around the river.

[6 marks]

08 . 2

Explain why decomposition of flooded vegetation in a new reservoir contributes to climate change.

[2 marks]

08 . 3

Explain how a river-regulation reservoir reduces both flooding and water shortages downstream.

[2 marks]

10

09

A group of students wants to investigate the effect of vegetation cover on rain-splash soil erosion.

09 . 1

Describe how they could carry out the investigation in the laboratory.

[5 marks]

09 . 2

Give **three** variables, other than those you have standardised, that affect rain-splash erosion in the field.

[3 marks]

1

2

3

09 . 3

Explain why it is easier to obtain valid results in the laboratory than in the field, and give **one** disadvantage of laboratory studies.

[2 marks]

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10

10 . 1

Evaluate the potential contribution of renewable energy resources to future energy supplies in a country such as the UK.

[9 marks]

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[illegible]

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10 . **2**

State the factor by which the kinetic energy of wind increases when wind velocity doubles.

[1 mark]

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10

Do not write outside the box

Write an essay on one of the following topics.

1	1	.	1
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Discuss how knowledge of the properties of pesticides can be used to reduce their environmental impacts while maintaining food production.

[25 marks]

OR

1 1 . 2

Discuss how knowledge of the properties of pollutants can be used to reduce the severity of their impact on the environment.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

[illegible]

Do not write outside the box

A large rectangular box containing 30 horizontal dotted lines for writing.

END OF QUESTIONS